



Lesson 4.8 — Unit 4 Review and Assessment

Big idea: Unit 4 is rational functions: where they're undefined (holes vs VAs), what they do at infinity (HA/slant), and how to combine them with all the operations of fractions.

Quick review

Holes vs. VAs: common factor cancels $\rightarrow$ hole; remaining denominator zero $\rightarrow$ VA.

End behavior: compare degrees. $n < d \rightarrow y = 0$; $n = d \rightarrow$ ratio of leading coefficients; $n = d + 1 \rightarrow$ slant via division.

Operations: simplify, multiply, divide, add, subtract — always factor first and find a common denominator for $\pm$.

Domain: exclude every x that makes any original denominator zero — even after cancellation.

Worked example

Worked review. $f(x) = (x^2 - 1)/(x^2 - x - 2) \rightarrow (x - 1)(x + 1)/[(x - 2)(x + 1)]$. Hole $x = -1$, VA $x = 2$, simplified $(x - 1)/(x - 2)$. HA $y = 1$.

Warm-ups

1. Find domain of $f(x) = (x - 1)/(x^2 - 9)$.
2. Find horizontal asymptote of $f(x) = (2x^2 + 1)/(3x^2 + 5)$.
3. Simplify $(x^2 - 16)/(x - 4)$.

Practice

1. Identify holes and VAs of $f(x) = (x - 3)/(x^2 - 9)$.
2. Find slant asymptote of $f(x) = (x^2 + 3x - 4)/(x - 1)$.
3. Compute $5/(x - 2) - 3/(x + 1)$.

Challenge

1. Sketch (describe) $f(x) = (x^2 - x - 2)/(x - 1)$ — intercepts, asymptotes, behavior on each interval.
2. Solve $[(x + 2)/(x - 1)] = [(x - 3)/(x - 1)] + 2$ for x .



Answer key — Lesson 4.8

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x \neq 3, -3$.
2. $y = 2/3$.
3. $(x - 4)(x + 4)/(x - 4) = x + 4$ ($x \neq 4$).

Practice

1. $(x - 3)/[(x - 3)(x + 3)] \rightarrow$ hole at $x = 3$, VA at $x = -3$.
2. Divide: $x + 4$ with remainder 0. Slant $y = x + 4$ (and actually no remainder $\Rightarrow$ exact equality, so the function equals $x + 4$ with hole at $x = 1$).
3. $[5(x + 1) - 3(x - 2)]/[(x - 2)(x + 1)] = (2x + 11)/[(x - 2)(x + 1)]$.

Challenge

1. Factor: $(x - 2)(x + 1)/(x - 1)$. VA $x = 1$. Slant: divide $(x^2 - x - 2)/(x - 1) = x + 0 + (-2)/(x - 1) \rightarrow$ slant $y = x$. y-intercept: 2. x-intercepts: 2 and -1. Sign chart breaks at -1, 1, 2.
2. Multiply both sides by $(x - 1)$: $x + 2 = x - 3 + 2(x - 1) = 3x - 5 \Rightarrow -2x = -7 \Rightarrow x = 7/2$ (must check $x \neq 1$: ok).